

Scaling Book Exercises: Section 2

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Source scan: ../handwritten/section_02_handwritten.pdf

Exercise 1

For v4p HBM bandwidth,

$$BW_{\text{HBM}} \approx 1.2 \times 10^{12} \text{ bytes/s.}$$

Moving 200GB in two directions gives

$$200\text{GB} \times 2 = 400 \times 10^9 \text{ bytes.}$$

Across 32 chips, this is

$$\frac{400 \times 10^9}{32} = 12.5 \times 10^9$$

bytes per chip. Therefore,

$$\text{total time} = \frac{12.5 \times 10^9}{1.2 \times 10^{12}} \approx 10^{-2} \text{ s} \approx 10 \text{ ms.}$$

Exercise 2

The scanned notes compare v5e and v5p hardware.

	v5e	v5p
cores	32	2240
TPU tensor cores	256	17920
total FLOPs/s	5.1×10^{16}	4.1×10^{18}
total HBM	4 TB	860 TB

For v5e, one calculation in the notes is

$$\frac{16 \cdot 16}{4 \cdot 2} = 32.$$

For v5p, the notes use:

- (a) $\frac{16 \cdot 20 \cdot 28}{2 \cdot 2 \cdot 1} = 4 \cdot 20 \cdot 28 = 2240,$
- (b) $2 \cdot 16 \cdot 20 \cdot 28 = 1120 \cdot 16 = 17920,$
- (c) $16 \cdot 20 \cdot 28 \cdot 4.59 \times 10^{14} \approx 4.0 \times 10^{18},$
- (d) $96 \cdot 16 \cdot 20 \cdot 28 = 860160 \text{ GB} \approx 860 \text{ TB.}$

Exercise 3

The computation time is

$$\text{FLOPs} = 2BDF, \quad \text{FLOPs/s}_{\text{v6e}} = 9.2 \times 10^{14},$$

so, for the scanned setting,

$$T_{\text{math}} = \frac{2BDF}{9.2 \times 10^{14}} = \frac{8BD^2}{9.2 \times 10^{14}}.$$

For memory communication, the notes use

$$2DF + 2BD + 2BF = 2(BD + DF + BF)$$

bytes and PCIe bandwidth

$$1.6 \times 10^{10} \text{ bytes/s.}$$

Thus,

$$T_{\text{comm}} = \frac{2(BD + DF + BF)}{1.6 \times 10^{10}}.$$

The compute-bound condition is

$$T_{\text{math}} > T_{\text{comm}}.$$

Using $F = 4D$ and $B \ll D$, this becomes

$$\frac{8BD^2}{9.2 \times 10^{14}} > \frac{8D^2}{1.6 \times 10^{10}},$$

and therefore

$$B > \frac{9.2 \times 10^{14}}{16 \times 10^9} = 5.75 \times 10^4 \approx 57500.$$

Exercise 4

Part (a)

For a matrix multiplication with dimensions shown in the scan,

$$\text{FLOPs} = 2 \cdot B \cdot 4096 \cdot 16384 = 134217728B \approx 1.3 \times 10^8 B.$$

Using v5e throughput,

$$\text{FLOPs/s}_{\text{v5e}} = 3.94 \times 10^{14},$$

the compute time is

$$T_{\text{math}} = \frac{1.3 \times 10^8}{3.94 \times 10^{14}} B.$$

The memory time is modeled as

$$T_{\text{mem}} = \frac{16384 \cdot 4096 + 4096B + 16384B}{8.2 \times 10^{11}} \approx \frac{6.7 \times 10^7 + 20000B}{8.2 \times 10^{11}}.$$

We are compute-bound if

$$T_{\text{math}} > T_{\text{mem}}.$$

Solving the inequality as in the scan gives

$$B > 267.$$

Part (b)

If loading is $22\times$ faster, then

$$T'_{\text{mem}} = \frac{T_{\text{mem}}}{22}.$$

Using the same inequality with the VMEM bandwidth denominator increased from 8.2×10^{11} to roughly 1.80×10^{13} , the compute-bound threshold becomes

$$B > 11.$$

In practice, since all VMEM bandwidth cannot usually be dedicated to this one load, the practical threshold is closer to 20.

Exercise 5

(a) With 3 rows and 3 columns, there are

$$3 + 3 = 6$$

links.

(b) The total bytes are

$$2 \cdot 8 \cdot 128 \cdot 8192 \approx 1.7 \times 10^7 \text{ bytes.}$$

The diagram in the notes routes from $(0,0)$ through $(0,1)$ and then to $(1,0)$, so the data can be sent two ways. With total bandwidth

$$2 \cdot 4.5 \times 10^{10} = 9.0 \times 10^{10},$$

the transfer time is

$$\frac{1.7 \times 10^7}{9.0 \times 10^{10}} \approx 188\mu\text{s}.$$

The notes record a total time of approximately

$$188\mu\text{s} + 6\mu\text{s} = 194\mu\text{s}.$$

Exercise 6

The data path in the notes is:

host DRAM $\rightarrow$ 16 TPU HBM shards $\rightarrow$ TPU $(0,0)$ HBM $\rightarrow$ MXU matrix multiplication.

The array is approximately

$$A \approx 17\text{GB}, \quad x = 2\text{MB}.$$

Each of the 16 TPUs holds approximately 1GB.

(i) Host DRAM to TPU HBM shards

Using PCIe bandwidth from Exercise 3,

$$\text{BW}_{\text{PCIe}} = 1.6 \times 10^{10} \text{ bytes/s} = 16\text{GB/s}.$$

With 16 PCIe lanes, the total bandwidth is approximately

$$16 \cdot 16 = 256\text{GB/s}.$$

Thus,

$$\text{time}_{\text{PCIe}} = \frac{16}{256} = \frac{1}{16} = 0.0625 \text{ s} = 62.5 \text{ ms}.$$

(ii) Receive remaining data from other TPUs

The (0,0) TPU must receive the remaining 15GB from other TPUs. Each link is approximately 45GB/s, so two links provide

$$90\text{GB/s}.$$

Therefore,

$$\text{time} = \frac{15}{90} = \frac{1}{6} \approx 0.167 \text{ s} = 167 \text{ ms}.$$

(iii) HBM to MXU

Moving 16GB from HBM to MXU with HBM bandwidth

$$8.2 \times 10^{11} \text{ bytes/s}$$

takes

$$\text{time} = \frac{16 \times 10^9}{8.2 \times 10^{11}} \approx 2 \times 10^{-2} \text{ s} = 20 \text{ ms}.$$

(iv) Compute

The compute required is

$$2 \cdot 8 \cdot (128 \cdot 1024)^2 = 2.75 \times 10^{11} \text{ FLOPs}.$$

Using throughput

$$1.97 \times 10^{14} \text{ FLOPs/s},$$

the compute time is

$$\frac{2.75 \times 10^{11}}{1.97 \times 10^{14}} = 1.4 \times 10^{-3} \text{ s} = 1.4 \text{ ms}.$$

The component times are

$$62.5 \text{ ms}, \quad 167 \text{ ms}, \quad 20 \text{ ms}, \quad 1.4 \text{ ms}.$$

Assuming perfect overlap and pipelining, the lower bound is

$$\max(62.5, 167, 20, 1.4) \text{ ms} = 167 \text{ ms}.$$

Verification Notes

The following items should be verified against the exercise text before final submission:

- The exact HBM bandwidth value in Exercise 1; the official text rounds the v4p value to about 1.2×10^{12} bytes/s and states a roughly 10 ms result.
- Exercise 5's topology labels and whether the $6\mu\text{s}$ latency term is per-hop or total.